DRUG REPOSITIONING FOR MUCOPOLYSACCHARIDOSES BASED ON SYSTEMS BIOLOGY DATA

Gerda Cristal Villalba Silva, Ursula Matte

Universidade Federal do Rio Grande do Sul | Bioinformatics Core | HCPA

INTRODUCTION

Mucopolysaccharidoses are lysosomal storage diseases.

In MPS, secondary cell disturbance affects pathways common to cancer.

AIMS

- To identify MPS-related drugs targeting oncogenic pathways
- To provide a list of candidate drugs for repurposing

METHODS

Gene expression analysis
edgeR v. 3.32 / limma v.3.46

Ontology analysis
ClueGO v.2.5.7

Network construcion
Stitch v.5.0

Drug search
Drugbank v.5.1

Network analysis
Cytoscape v.3.8.2

Interaction network
Omnipath v.2.0

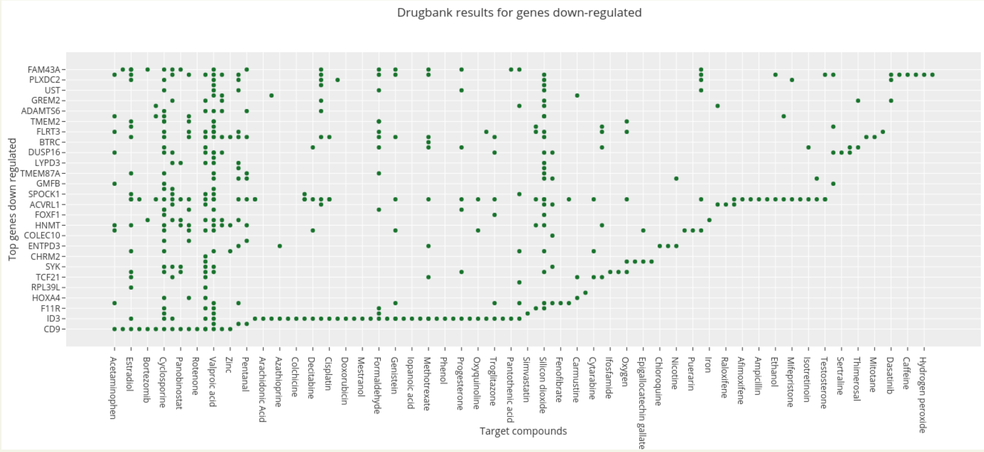
Gene hub analysis
Cytohubba v. (MCC method)

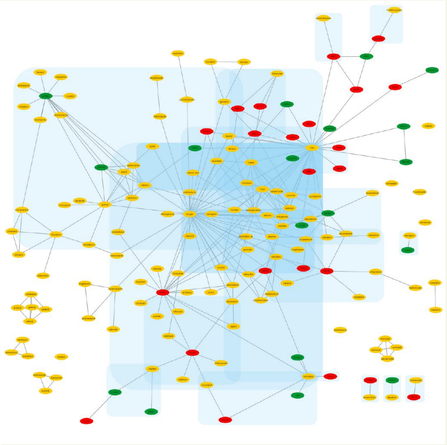
Datasets available at

MPSbase

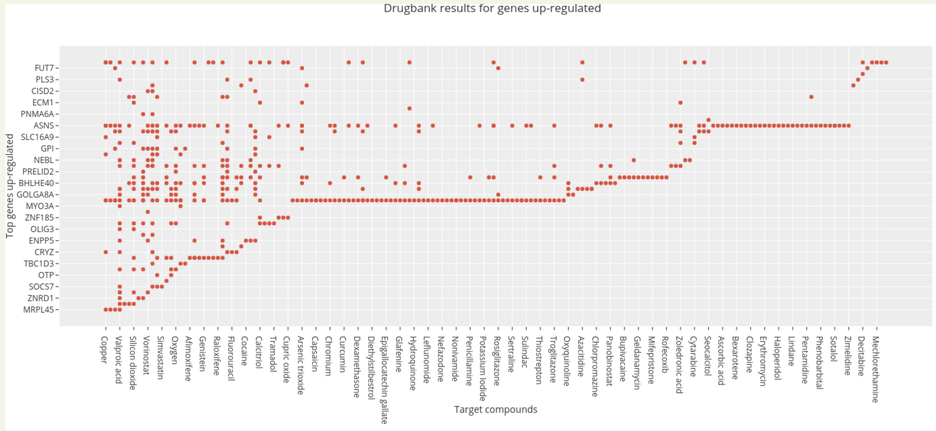
HTTPS://WWW.UFRGS.BR/MPSBASE/

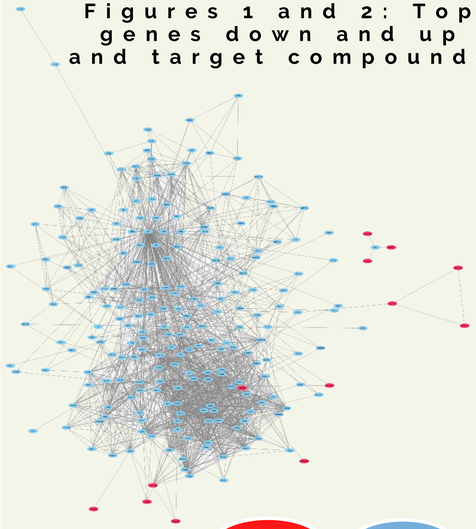
RESULTS





Figures 1 and 2: Top genes down and up and target compounds





drugs genes

Figure 3: Network by cytoscape nodes: 244 edges: 1824.

Gene	Status	compound	Pubchem
BTRC	down	Methotrexate results in increased expression of BTRC mRNA	https://pubchem.ncbi.nlm.nih.gov/compound/126941
CGB7	Up	Seocalcitol results in decreased expression of CGB7 mRNA	https://pubchem.ncbi.nlm.nih.gov/compound/14829
DNAJC8	Down	Valproic Acid results in increased expression of DNAJC8 mRNA	https://pubchem.ncbi.nlm.nih.gov/compound/3121
HSD17B7P2	Down	Zoledronic acid results in increased expression of HSD17B7P2 mRNA	https://pubchem.ncbi.nlm.nih.gov/compound/68740
SLITRK2	Down	Gemcitabine results in increased expression of SLITRK2 mRNA	https://pubchem.ncbi.nlm.nih.gov/compound/60750

Drugbank genes with unique match and opposite expression

Table 1: Genes with unique compounds

DISCUSSION

The 51 genes down-regulated o the network participate in several processes like regulation of platelet aggregation and regulation, cell adhesion, cellular response to low-density lipoprotein particle stimulus, and phosphorylation.

The 44 genes up-regulated participate in kinase and intramolecular oxidoreductase pathways, and other metabolic signaling pathways.

Despite the compounds, most of them are Antineoplastic and Immunomodulating Agents, inhibitors, or proteasome targets.

CONCLUSIONS

Our results suggest that drugs modulating the Axon guidance, EGFR, mTOR, Wnt, and immune system pathways, are particularly promising for intervention. Furthermore, the list of drugs and related MPS enriched genes could be useful not only as new treatments but also considered for pathophysiological studies.

FINANTIAL SUPPORT

HOSPITAL DE CLÍNICAS PORTO ALEGRE RS

CNPq